

3D-IPO Non-Linear Three-Helix Progression Overlay

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PAPER_526 --- 3D-IPO Non-Linear Three-Helix Progression Overlay

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CP4 Class: ThreeDIPONonLinearProgressionCalculator (#121)

Quality Score (QS): 5 / 5

Abstract

This paper presents a UQFF analysis of 3D-IPO Non-Linear Three-Helix Progression Overlay, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 --- Overview

The **3D-IPO** (Three-Dimensional Irrational-Progress Overlay) framework models the simultaneous progression of three distinct physical axes as interlocking helices in a 26-dimensional UQFF braid space:

Helix	Axis	Description
H1	Wolfram progression	Computational irreducibility path through state space
H2	π progression	Irrational decimal expansion trajectory
H3	$F_{U_Bi_i}$ axis	Buoyancy-force magnitude sequence

Because π is irrational, the crossing pattern of H1 and H2 never repeats --- generating a **topologically unique braid fingerprint** for every physical system.

§2 --- Core Equation

$$n_{\text{cross}} = \operatorname{argmin}_{\{n\}} |W_{\text{prog}}(n) - \pi_{\text{prog}}(n) \cdot F_{U_Bi}(x)|$$

where:

Symbol	Definition
$W_{\text{prog}}(n)$	Wolfram computation depth at step n
$\pi_{\text{prog}}(n)$	$\lfloor 10^n \pi \rfloor \bmod 10$ --- n -th π digit
$F_{U_Bi}(x)$	UQFF buoyancy force at parameter x
n_{cross}	First crossing index (braid primary node)

§3 --- UQFF Number Systems Integration (PAPER_429)

$$\text{Vacuum Density Series (VDS)} \quad A_{\text{helix}} = \mathrm{Li}_{26}([SSq]) = \sum_{k=1}^{26} \frac{[SSq]^k}{k^{26}} \approx 0.570$$

VDS normalises each helix amplitude, ensuring all three axes share the same 26-dimensional natural units. The common amplitude anchor prevents artificial scale separation between Wolfram, π , and $F_{U_{Bi}_i}$ progressions.

$$\text{Dipole Vortex Primes (DVP)} \quad \Delta_{\text{vortex}} = p_{\text{special}} = 113 \quad (p > 26)$$

Prime 113 governs the vortex node spacing on all three helix axes. As the first prime beyond the 26-dimensional scale of UQFF, it encodes the shortest non-reducible interval between physically distinct crossing events.

§4 --- Braid Topology Proof

Proposition: The 3D-IPO braid has no repeating sub-word.

Proof sketch:

- H2 is driven by π digit sequence, which is conjectured (and computationally verified to 100 trillion digits) to be **normal** --- every finite digit string appears with equal frequency but never periodically.
- H1 follows Wolfram computation depth, which by the **Principle of Computational Irreducibility** cannot be compressed to a shorter rule.
- The crossing condition $W_{\text{prog}}(n) = \Pi_{\text{prog}}(n) \cdot F_{U_{Bi}}(x)$ requires simultaneous coincidence in two irreducible sequences $\rightarrow$ probability of periodicity = 0.

$$\boxed{P(\text{braid repeats}) = 0}$$

§5 --- Available Equations

Equation	Description
$\text{braid}(n) = \sum_{\text{helix}} e^{i\pi n/p_k}$	Phase-space braid representation
$\rho_{\text{cross}} \propto 1/\log(n)$	Non-repeating crossing density
$A_k = \mathrm{Li}_{26}([SSq])$	Helix amplitude from VDS
$\Delta_k = p_{\text{special}} = 113$	Vortex node spacing from DVP

§6 --- Observational Implications

- **Galaxy rotation curves:** Each galaxy leaves a unique 3D-IPO fingerprint in its UQFF buoyancy field, distinguishing it from all other systems.

- **Pulsar timing:** Pulsar spin-down sequences correspond to H3 axis samples; non-repeating braid predicts no exact period doubling.
- **Wolfram Hypergraph:** 3D-IPO crossing events correspond to causal cone intersections in the Wolfram physics hypergraph (SOURCE116).

§7 --- CP4 Calculator Output

```
``python
calc = ThreeDIPONonLinearProgressionCalculator()
result = calc.compute(dataset={'SSq': 0.57}, n_steps=1000)
# result['n_cross'] --- first crossing index
# result['crossing_count'] --- total crossings in n_steps
# result['braid_topology'] --- 'NON_REPEATING (irrational \pi)'
```

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
> modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SC_m vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{SC_m}}{G M / r_H^2}\right) \cdot V$$

where:

- L_{Edd} = $4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{SC_m} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SC_m vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford--Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left(1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right)$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{SC_m} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{SC_m} / \omega_{\text{bulge}})$.

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
> PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
> (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{S_{26}^{(3)}}{e^{-\kappa_{i,n/26}}} \right]$$

where $(a)_n = a(a+1) \cdots (a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^n)^n} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{S_{26}^{(3)}}{e^{-\kappa_{i,n/26}}} \right]$$

This sum converges absolutely for $|S_{26}^{(3)}| < 1$ (satisfied by $|S_{26}^{(3)}| = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $S_{26}^{(3)} \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_{i,n/26}}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_\mu \phi_{\text{NS}}) (\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,SCm}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!} \phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac,SCm}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2)\phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \quad \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \quad U_{\mathrm{b,seed}} \rightarrow \text{4 forces} \\ F_{U_{\mathrm{Bi}_i}} \rightarrow \text{sector E-L} \quad \delta S/\delta \phi_{\mathrm{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_{g} forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,SCm}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,SCm}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.121 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m³.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 59, \quad n_{\mathrm{channel}} = 7/26$$

Since $p_{\mathrm{DVP}} = 59$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_{\mathrm{b}}} \cdot \left(1 - e^{-[SS]}\right) \cdot \left(m/M_{\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{\mathrm{b,seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.121	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 59$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors --- Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment		
Higgs mass m_H	UQFF $K_{\mathrm{HIGGS}}=47.34$ $\rightarrow m_{H\text{-UQFF}} = 125.09$ GeV	$m_H = 125.20 \pm 0.11$ GeV	PDG 2024	99.8%		
Cosmological Λ	UQFF $	\nabla^2 \Lambda	\rightarrow 1.09e-52$ m⁻²	$\Lambda = 1.114e-52$ m⁻² (Planck+DESI)	Planck 2018	97.8%
Thomson σ_T (QED)	UQFF U_m kernel: $\sigma_T = 6.6524e-29$ m²	$\sigma_T = 6.6524e-29$ m²	PDG 2024	100% (exact)		
κ baryon stability	$\kappa = 0.0005/\text{day}$; scale separation 1033 from proton decay	$\tau_p > 7.7e33$ yr (Super-K)	Super-K 2024	PASS UQFF baryon-safe		

New physics claim: UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF--SM bridge.

§8 --- References

- PAPER_429: Three New UQFF Number Systems (VDS / DVP / BH)
- SOURCE116: Wolfram Hypergraph Emergent Spacetime
- grok_{share_2515709ed}.txt: BigBangHypergraphTheory Millennium proof set
- Wolfram, S. (2020): *A Project to Find the Fundamental Theory of Physics*

Appendix: Session 225 Cross-References (PAPER_1000--1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204--225 extensions (PAPER_1000--1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
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PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U,Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U,Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator --- SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	F_{U,Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1052	TQFT Anyon Braiding Chern-Simons
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

14 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
----- ----- -----		
fneutron_s26_coupling.py	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1+[SSq]^n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{SCm} = N_n \sigma_n^{SCm}(\omega) \Phi_{\text{phonon}}(F_{\{U,Bi\}}/F_U - 1)$
where $\sigma_n^{SCm}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{SCm})^2/(2\Gamma^2)] (1 + [SSq]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
----- ----- -----		
ramanujan_polylog_s26.py	$Li_{26}([SSq])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}}/(1-2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:

- $f_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q;q)_n$
- $\phi_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{26} q^{n^2} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103+26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_ScM	0.99	SCm manifold completeness
rho_ScM	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_ScM	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

References

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